

AZ-700

Design and implement Azure ExpressRoute



Module Overview

- Explore Azure ExpressRoute
- Design an ExpressRoute deployment
- Exercise – Configure an ExpressRoute Gateway
- Exercise – Provision an ExpressRoute circuit
- Configure peering for an ExpressRoute deployment
- Connect an ExpressRoute circuit to a VNet
- Connect geographically dispersed networks with ExpressRoute Global Reach
- Improve data path performance between networks with ExpressRoute FastPath
- Troubleshoot ExpressRoute connection issues

Explore Azure ExpressRoute



Learning Objectives – Azure ExpressRoute

- ExpressRoute Capabilities
- Understand use cases for Azure ExpressRoute
- ExpressRoute connectivity models
- Design considerations for ExpressRoute deployments
- Bidirectional Forwarding Detection (BFD)
- Configure encryption over ExpressRoute
- Learning Recap

ExpressRoute Capabilities

Layer 3 connectivity with redundancy

Connectivity to all regions within a geography

Global connectivity with ExpressRoute premium add-on

Across on-premises connectivity with ExpressRoute Global Reach

Bandwidth options – 50 Mbps to 100 Gbps

Billing models – Unlimited, metered



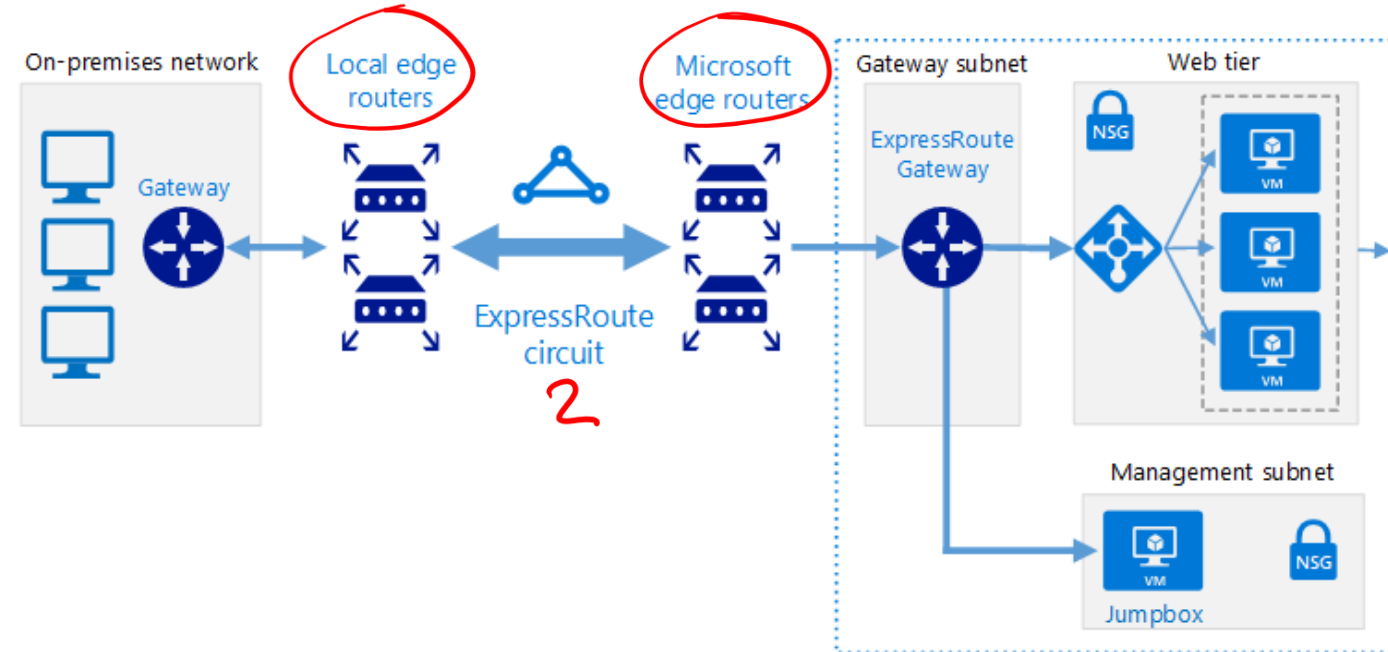
Understand use cases for Azure ExpressRoute

Benefits

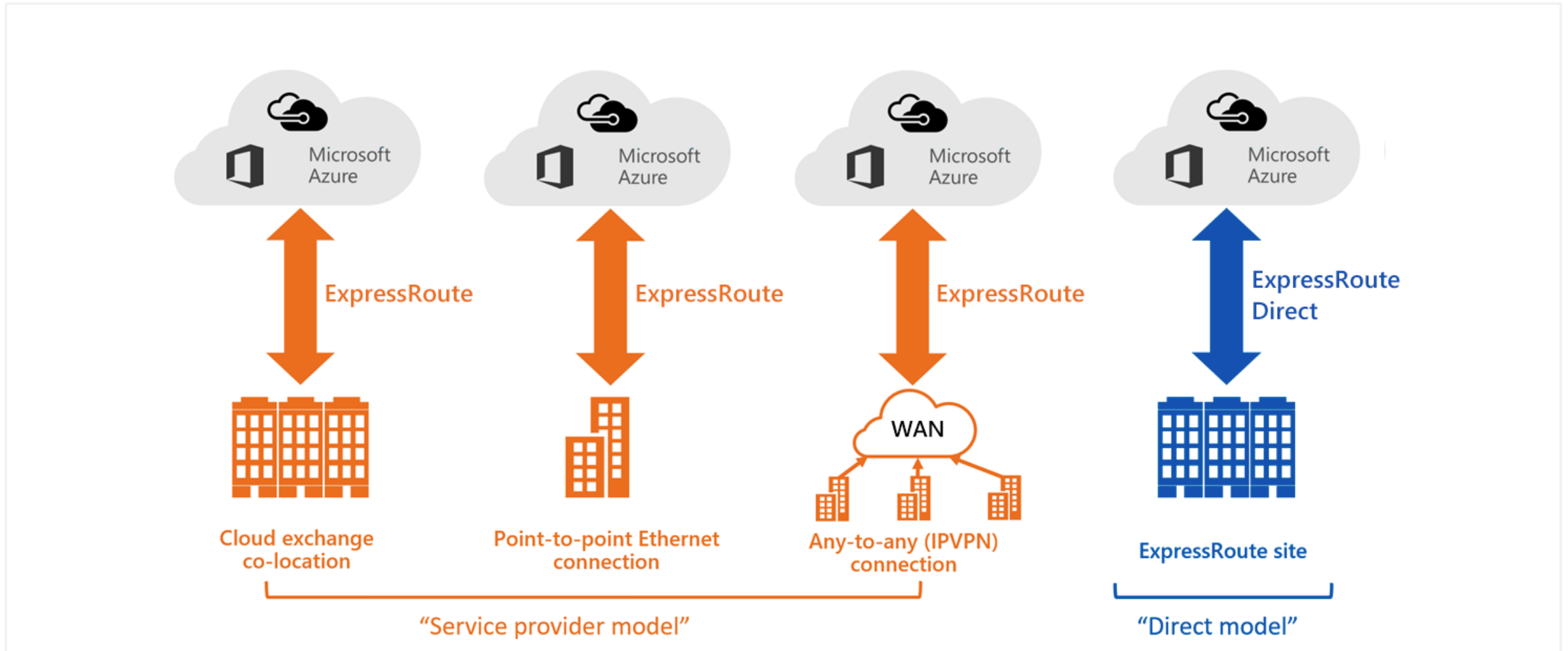
- Predictable, reliable, and high-throughput connections
- Up to 100 Gbps bandwidth - supports dynamic scaling of bandwidth and direct access to national clouds
- Built in redundant circuits
- Integrates with existing Multiprotocol Label Switching (MPLS)
- Up to 99.95% availability SLA across the entire connection.

Challenges

- Working with a third-party connectivity provider
- Requires high-bandwidth routers on-premises



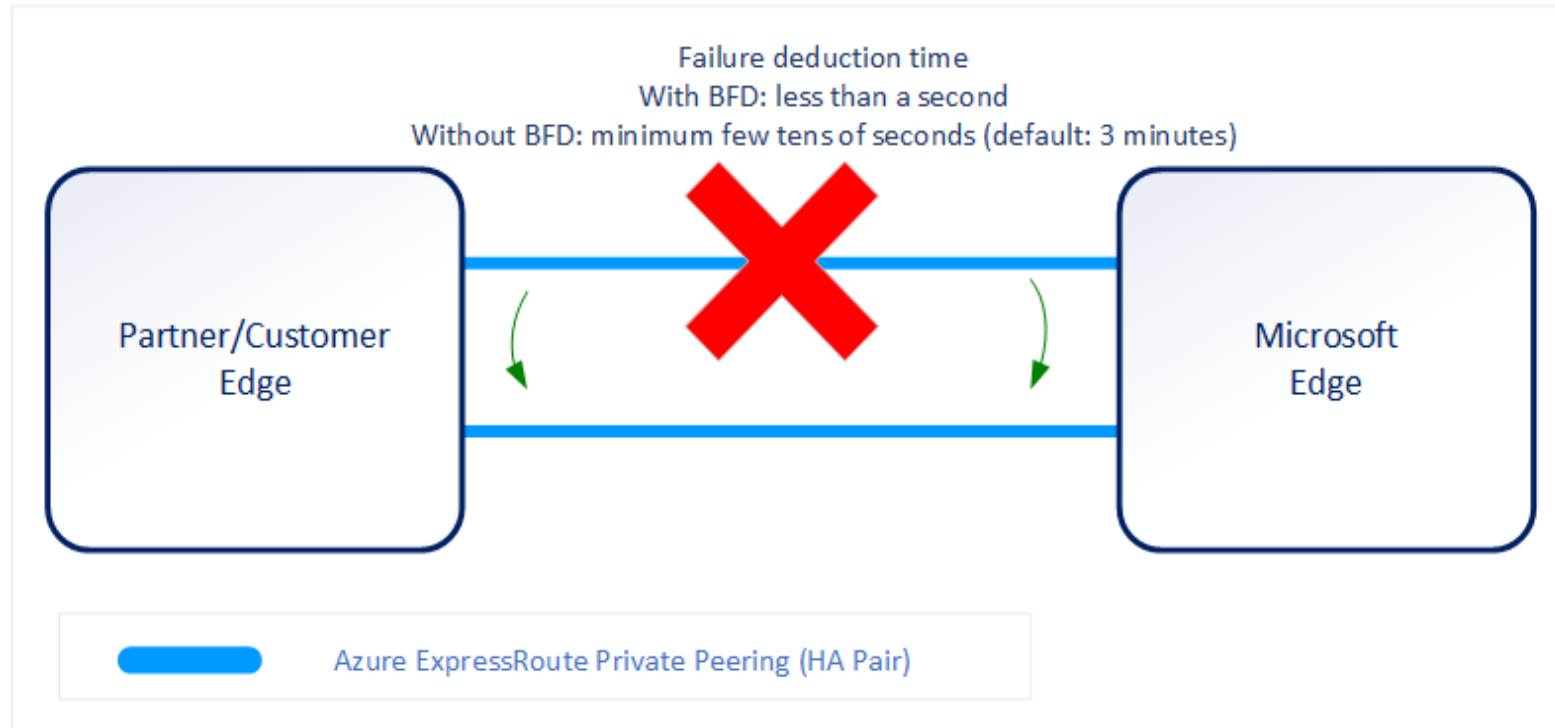
ExpressRoute connectivity models



Design considerations for ExpressRoute deployments

ExpressRoute using a Service Provider	ExpressRoute Direct
Uses service providers to enable fast onboarding and connectivity into existing infrastructure	Requires 100 Gbps/10 Gbps infrastructure and full management of all layers
Integrates with hundreds of providers including Ethernet and MPLS	Direct/Dedicated capacity for regulated industries and massive data ingestion
Circuits SKUs from 50 Mbps to 10 Gbps	Customer may select a combination of the following circuit SKUs on 100-Gbps ExpressRoute Direct: (5 Gbps, 10 Gbps, 40 Gbps, 100 Gbps) Customer may select a combination of the following circuit SKUs on 10-Gbps ExpressRoute Direct: (1 Gbps, 2 Gbps, 5 Gbps, 10 Gbps)
Optimized for single tenant	Optimized for single tenant with multiple business units and multiple work environments

Bidirectional Forwarding Detection (BFD) with ExpressRoute Peering

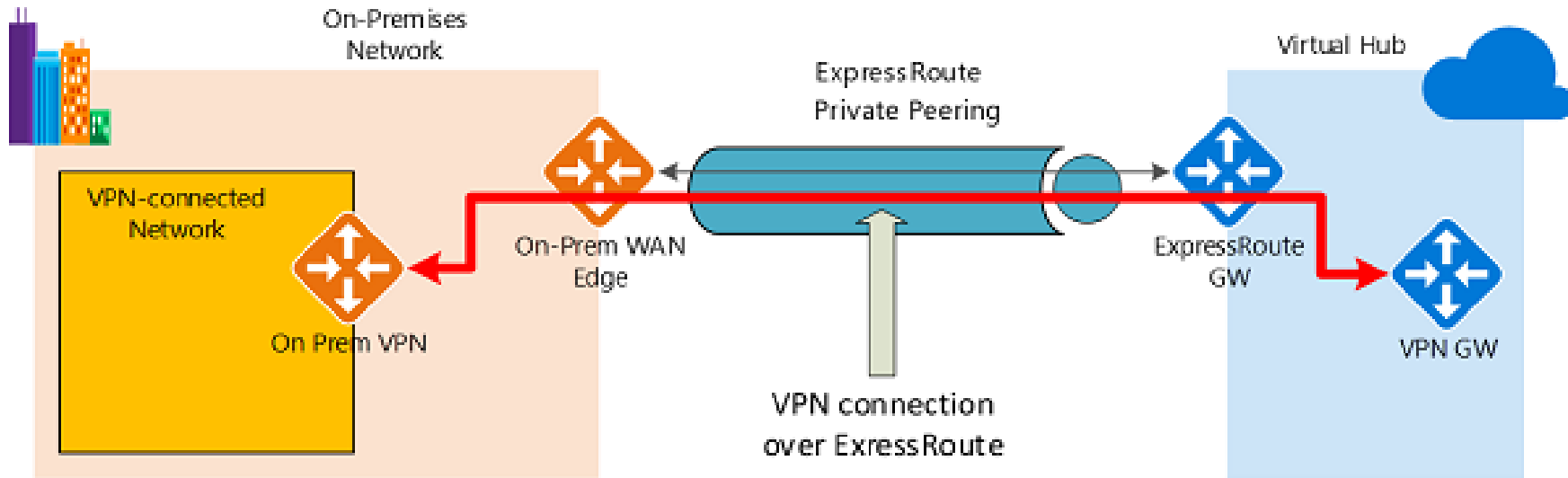


BFD is configured by default on the Microsoft Edge

You only need to configure BFD on both your primary and secondary devices

You configure the BFD on the interface and then link it to the BGP session.

Configure encryption over ExpressRoute

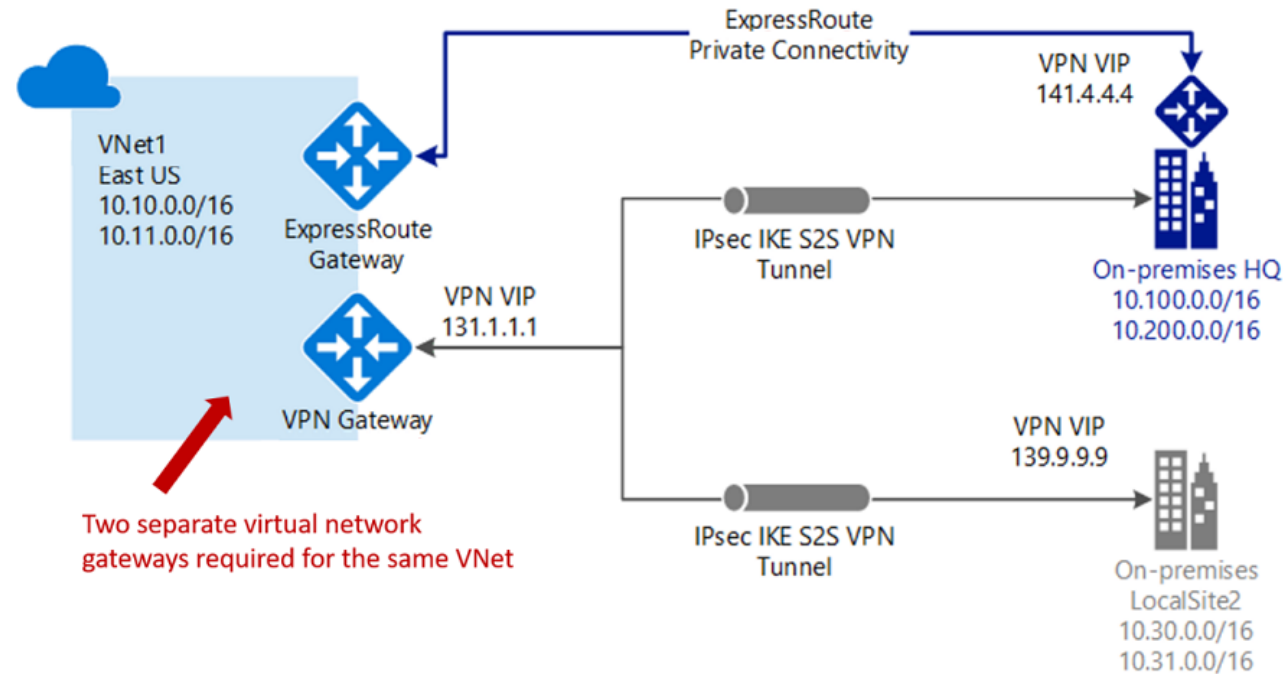


Establish ExpressRoute connectivity with an ExpressRoute circuit and private peering

Establish the VPN connectivity over ExpressRoute

Routing between the on-premises networks and Azure over both the ExpressRoute and VPN paths

Coexisting Site-to-Site and ExpressRoute



Use S2S VPN as a secure failover path for ExpressRoute

Use S2S VPNs to connect to sites that are not connected with ExpressRoute

Notice two VNet gateways for the same virtual network

Design an ExpressRoute deployment



Learning Objectives – Design an ExpressRoute Deployment

- ExpressRoute SKUs and Tiers
- Choose a peering location
- Choose the right ExpressRoute Circuit and billing model
- Demonstration
- Learning Recap

ExpressRoute SKUs

Local (if available) - provides free egress data transfer and gives you access to only 1-2 Azure regions in the same area as your circuit

Standard SKU - gives you access to all Azure regions in a geopolitical area

Premium SKU - provides support for more than 4K routes, ability to connect to more than 10 virtual networks, and global connectivity

Basics Configuration Tags Review + create

ExpressRoute circuits can connect to Azure through a service provider or directly to Azure at a global peering location. [Learn more about circuit types](#)

Port type * ⓘ ☒ Provider ☐ Direct

Create new or import from classic * ⓘ ☒ Create new ☐ Import

Provider * ⓘ

SKU * ⓘ ☐ Local ☒ Standard ☐ Premium

Billing model * ⓘ ☒ Metered ☐ Unlimited

Allow classic operations ⓘ ☐ Yes ☒ No

Choose a peering location



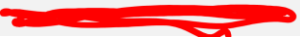
Choose the right ExpressRoute Circuit and billing model

Choose Metered or unlimited data plan

Choose Bandwidth

You can increase gateway size but not decrease without service outage

Pricing varies by region and zone



Unlimited data. Billing is based on a monthly fee; all inbound and outbound data transfer is included free of charge.

Metered data. Billing is based on a monthly fee; all inbound data transfer is free of charge. Outbound data transfer is charged per GB of data transfer. Data transfer rates vary by region.

ExpressRoute premium add-on. ExpressRoute premium is an add-on to the ExpressRoute circuit.

Configure peering for an ExpressRoute deployment



Learning Objectives - Peering for an ExpressRoute

- Configure Private peering
- Configure Microsoft peering
- Choose between private peering only, Microsoft peering only, or both
- Route filters
- Learning Recap

Configure Private peering

A pair of subnets that are not part of any address space reserved for virtual networks. One subnet will be used for the primary link, while the other will be used for the secondary link

A valid on-prem VLAN ID to establish this peering on

AS number for peering

Advertise the routes from your on-premises Edge router to Azure via BGP

Optional - An MD5 hash if you choose to use one

 **Private peering**  

ASH-Cust15-ER

Peer ASN * ⓘ

65021 ✓

Subnets

☒ Both

☐ IPv4

☐ IPv6

IPv4 Primary subnet * ⓘ

192.168.15.16/30 ✓

IPv4 Secondary subnet * ⓘ

192.168.15.20/30 ✓

☒ Enable IPv4 Peering ⓘ

IPv6 Primary subnet * ⓘ

fd:1:2:15FF::/126 ✓

IPv6 Secondary subnet * ⓘ

fd:1:2:15FF::4/126 ✓

☒ Enable IPv6 Peering ⓘ

VLAN ID * ⓘ

150 ✓

Shared key

Add Global Reach

Global Reach name

ExpressRoute Circuit name ⓘ

IPv4 Subnet ⓘ

IPv6 Subnet ⓘ

Save

Cancel

Configure Microsoft peering

A pair of subnets owned by you and registered in an RIR/IRR. One subnet will be used for the primary link, while the other will be used for the secondary link.

A valid on-prem VLAN ID to establish this peering on

AS number for peering

Advertised prefixes: You provide a list of all prefixes you plan to advertise over the BGP session

Routing Registry Name

[Home](#) > [ExpressRoute circuits](#) > [TestERCircuit](#) >

 **Microsoft peering**  
TestERCircuit

 To receive route advertisements on Microsoft peering, attach route filters to the circuit after creating Microsoft Peering. [Learn More.](#)

☒ Enable Peering 

Peer ASN * 
 

IPv4 Primary subnet * 
 

IPv4 Secondary subnet * 
 

VLAN ID * 
 

Customer ASN 
 

Routing registry name 
 

Shared key

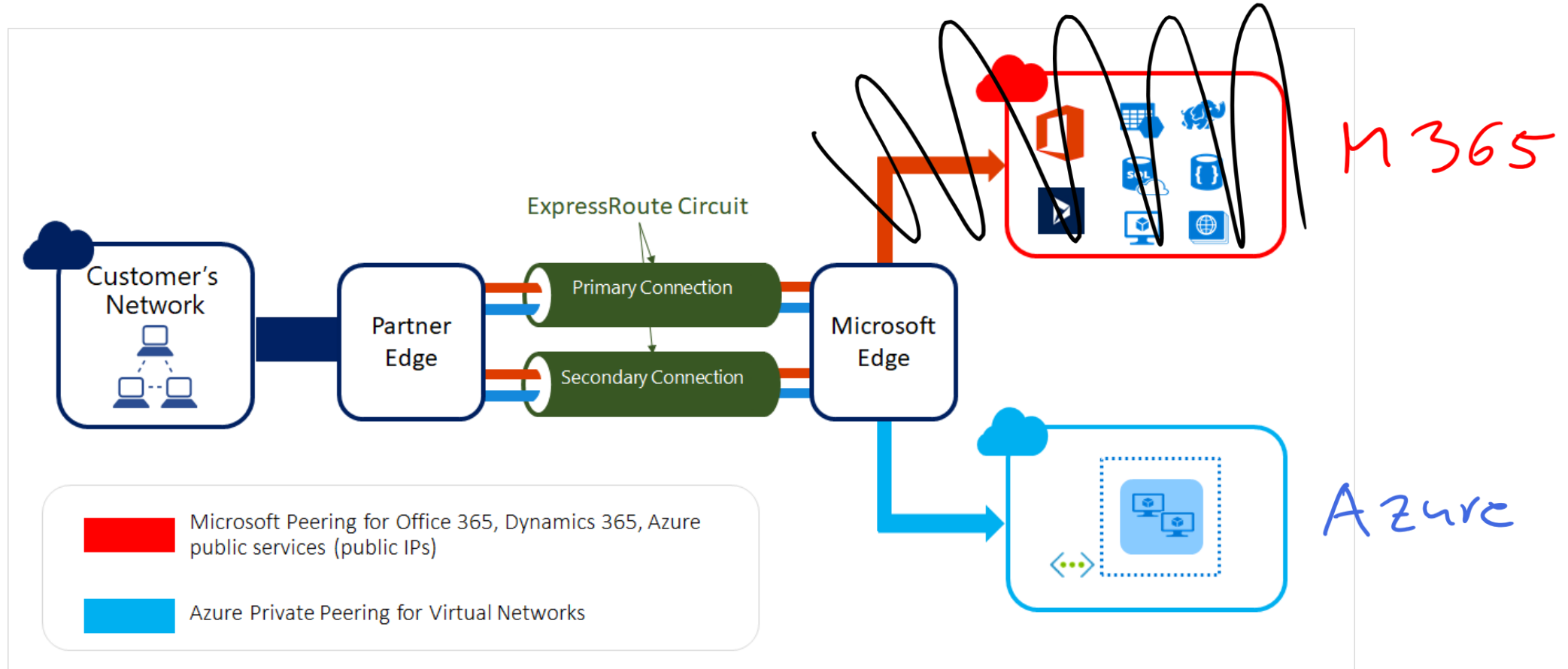
Advertised public prefixes * 
 

Status: Validation Needed

Save

Cancel

Choose between private peering only, Microsoft peering only, or both



Route filters

Create route filter

Basics Tags Review + create

Route filters help you filter the traffic going through your ExpressRoute. Essentially a white list of all the BGP community values, a route filter lets you identify services you want to consume through your ExpressRoute circuit's Microsoft peering. Once a route filter resource is defined and attached to an ExpressRoute circuit, all prefixes that map to the BGP community values are advertised to your network. [Learn more about route filters](#)

Project details

Subscription * ① Azure Subscription

Resource group * ① ExpressRouteResourceGroup
[Create new](#)

Instance details

Name * MyRouteFilter ✓

Region * (US) West US 2 ✓

i Route filter must be created in the same location as the ExpressRoutes it will be associated with.

[Review + create](#)

[< Previous](#)

[Next : Tags >](#)

[Download a template for automation](#)

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Manage rule

MyRouteFilter

[Save](#) [Discard](#)

Rule name * Rule1 ✓

Allowed service communities * 2 selected

- ☒ Select all
- ☐ Exchange (12076:5010)
- ☒ Other Office 365 Services (12076:510)
- ☒ SharePoint Online (12076:5020)
- ☐ Skype For Business (12076:5030)
- ☐ CRM Online (12076:5040)
- ☐ Azure Active Directory (12076:5060)
- ☐ Azure Australia Central (12076:5103)
- ☐ Azure Australia Central 2 (12076:510)
- ☐ Azure Australia East (12076:51015)
- ☐ Azure Australia Southeast (12076:51)
- ☐ Azure Brazil South (12076:51014)
- ☐ Azure Canada Central (12076:51020)
- ☐ Azure Canada East (12076:51021)
- ☐ Azure Central India (12076:51017)
- ☐ Azure Central US (12076:51009)
- ☐ Azure Central US EUAP (12076:5100)
- ☐ Azure East Asia (12076:51010)
- ☐ Azure East US (12076:51004)
- ☐ Azure East US 2 (12076:51005)

Create route filter and
create route filter rules

Attach the route filter to
an ExpressRoute circuit

Connect an ExpressRoute circuit to a VNet



Learning Objectives - Connect an ExpressRoute Circuit to a VNet

- Connect virtual network to an ExpressRoute circuit
- Learning Recap

Connect virtual network to an ExpressRoute circuit



TestERCircuit | Connections

ExpressRoute circuit

Search (Ctrl+ /)

<< **+ Add** Delete Refresh

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Settings

Configuration

Connections

Authorizations

Search connections

Name ↑↓	Status ↑↓	Connection type ↑↓	Peer ↑↓
No results.			

Connect virtual network to an ExpressRoute circuit – continued

Create connection

Basics

Settings

Tags

Review + create

Create a secure connection to your virtual network by using VPN Gateway or ExpressRoute. [Learn more about VPN Gateway](#)
[Learn more about ExpressRoute](#)

Project details

Subscription

Azure Subscription

Resource group

ExpressRouteResourceGroup

Create new

Instance details

Connection type

ExpressRoute

Name

ER-VNet-Connection

Region

(US) West US 2

Review + create

< Previous

Next : Settings >

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Create connection

Basics

Settings

Tags

Review + create

Virtual network gateway

To use a virtual network with a connection, it must be associated to a virtual network gateway.

Virtual network gateway

ERGW

ExpressRoute circuit

TestERCircuit

Redeem authorization

☐

Routing weight

0

Review + create

< Previous

Next : Tags >

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Verify connection

 **TestERCircuit** | Connections 

ExpressRoute circuit

<<

 Add

 Delete

 Refresh

 Overview

 Activity log

 Access control (IAM)

 Tags

 Diagnose and solve problems

Settings

 Configuration

 Connections

 Authorizations

Name 	Status 	Connection type 	Peer 
ER-VNet-Connection	Succeeded	ExpressRoute	ERGW

Connect geographically dispersed networks with ExpressRoute Global Reach



Learning Objectives - ExpressRoute Global Reach

- Use Cross-region connectivity to link multiple ExpressRoutes
- Choose when to use ExpressRoute Global Reach
- Configure ExpressRoute Global Reach
- Learning Recap

Use Cross-region connectivity to link multiple ExpressRoutes

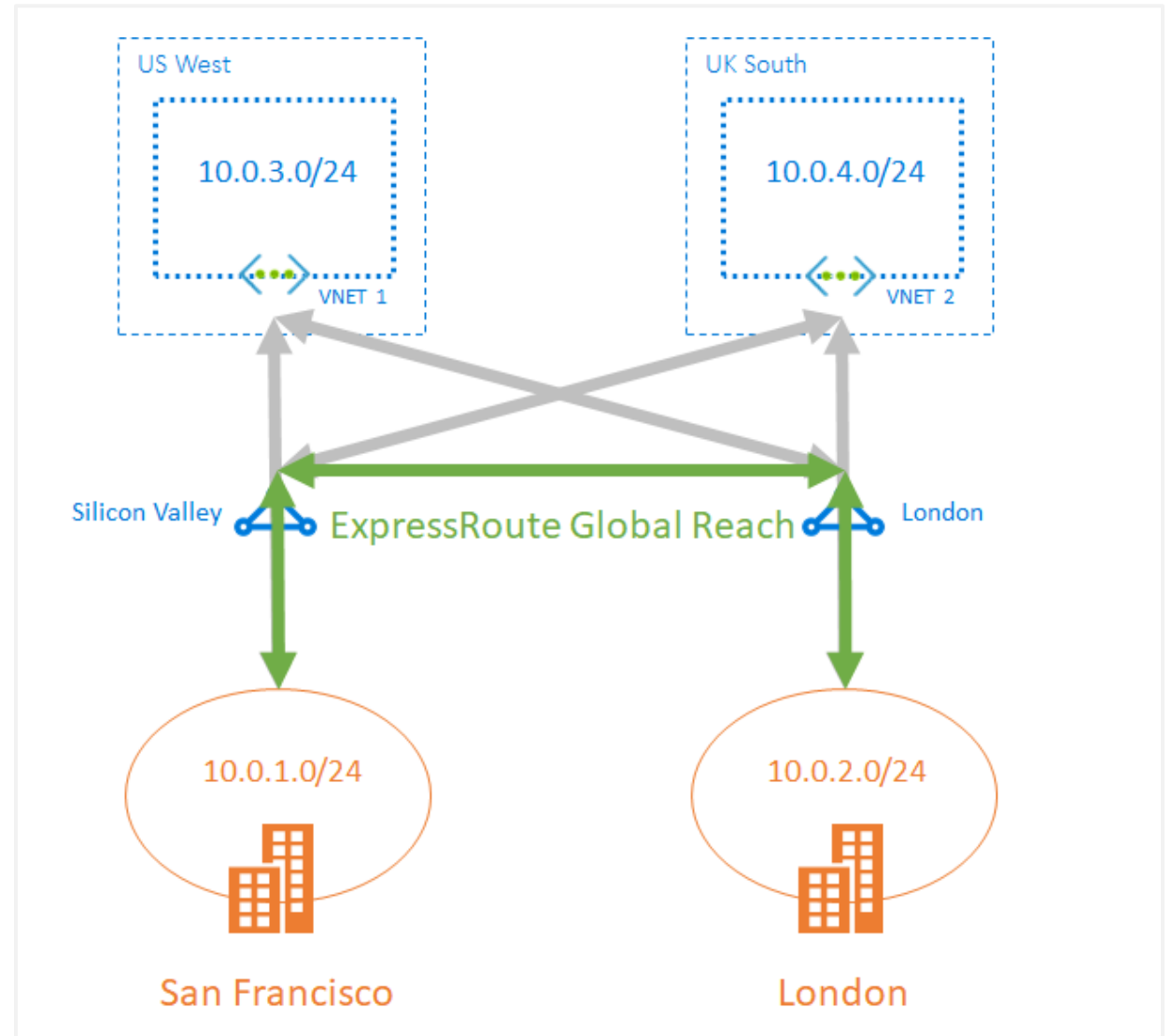
Connectivity to all regions within a geopolitical region

Global connectivity with ExpressRoute Premium

Local connectivity with ExpressRoute Local

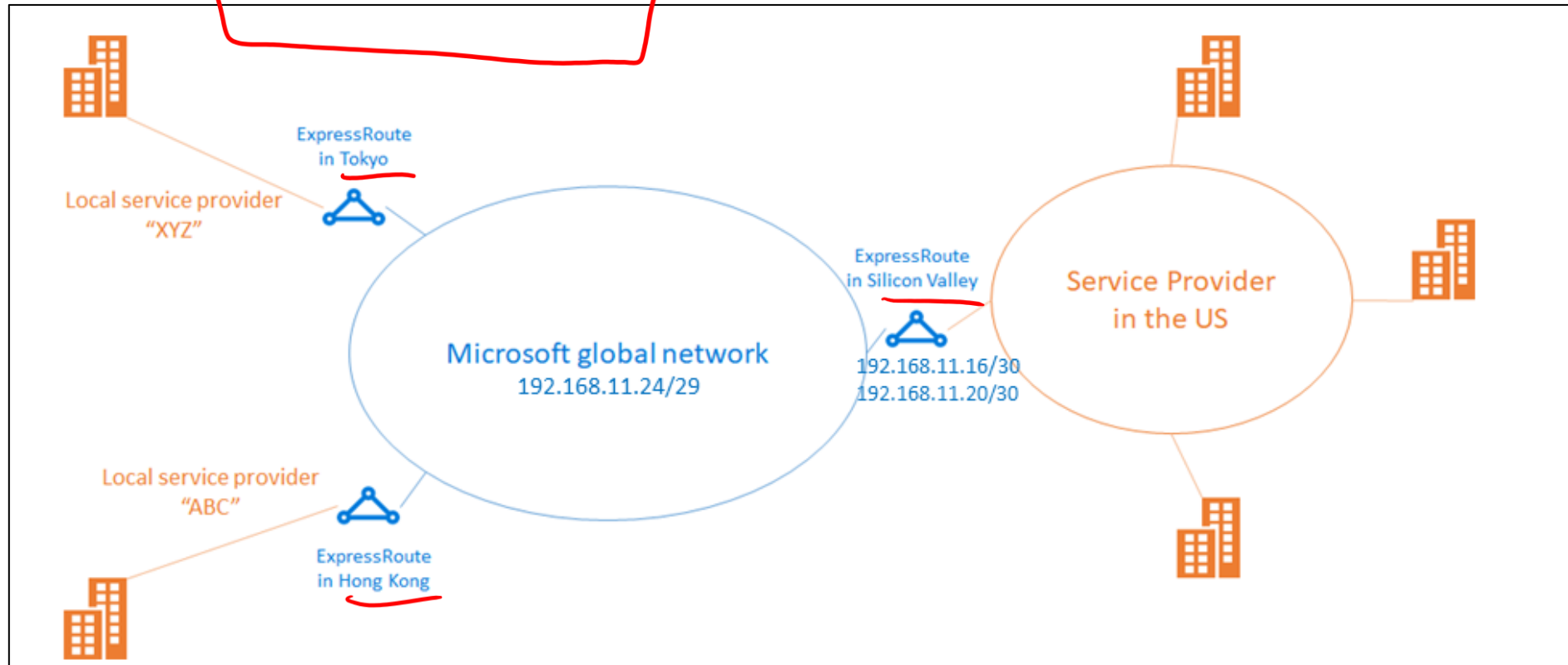
Across on-premises connectivity with ExpressRoute Global Reach

ExpressRoute Direct



ExpressRoute Global Reach

UWAN



Designed to complement your service provider's WAN implementation and connect your branch offices across the world

You can link ExpressRoute circuits together to make a private network between your on-premises networks

Configure ExpressRoute Global Reach

Private peering 

TestERCircuit

☒ Enable Peering ⓘ

Peer ASN * ⓘ
65020 ✓

IPv4 Primary subnet * ⓘ
192.168.11.16/30

IPv4 Secondary subnet * ⓘ
192.168.11.20/30

VLAN ID * ⓘ
110 ✓

Shared key

Add Global Reach

Global Reach name	ExpressRoute Circuit name ⓘ	Global Reach subnet ⓘ

Save

Cancel

Add Global Reach 

TestERCircuit

Global Reach name *
TestERCircuit1-TestERCircuit2 ✓

☐ Redeem authorization ⓘ

ExpressRoute circuit * ⓘ
TestERCircuit2
resourceGroup: ExpressRouteResourceGroup2, location: eastus ✓

Global Reach subnet * ⓘ
192.168.11.24/29 ✓

Add

Cancel

Improve data path performance between networks with ExpressRoute FastPath



Learning Objectives - ExpressRoute FastPath

- ExpressRoute FastPath
- Configure ExpressRoute FastPath
- Learning Recap

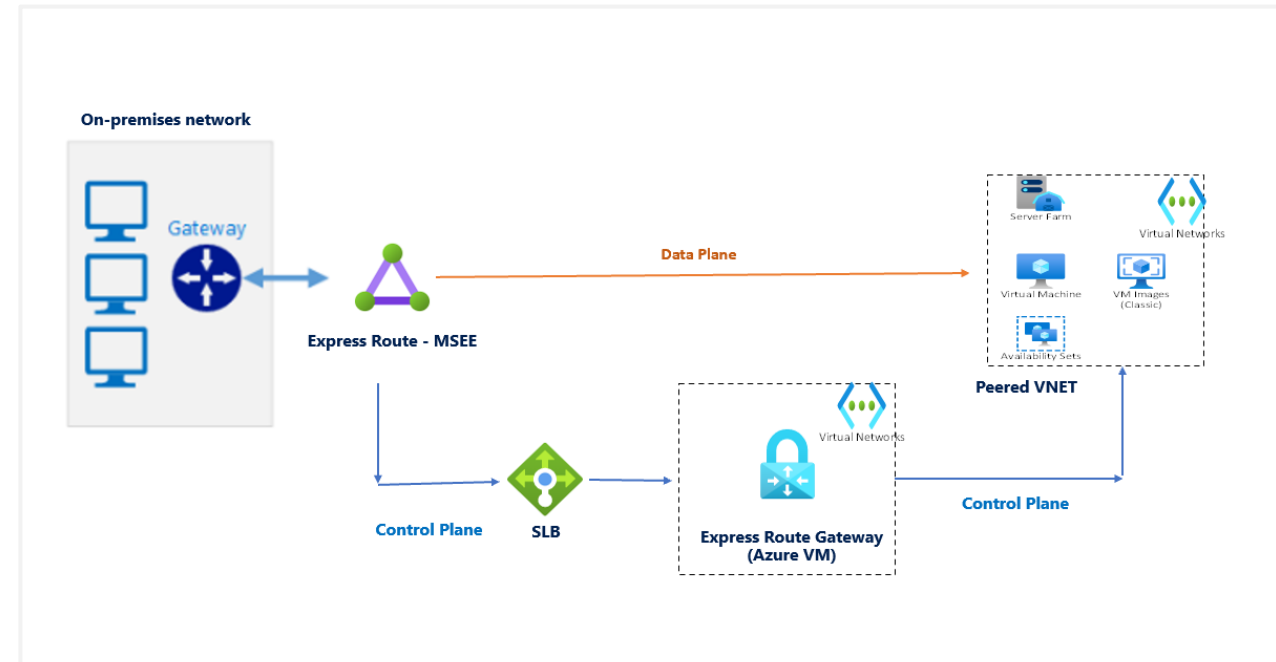
ExpressRoute FastPath

FastPath is designed to improve the data path performance between your on-premises network and your virtual network.

When enabled, FastPath sends network traffic directly to virtual machines in the virtual network, bypassing the gateway.

FastPath improves data path performance such as packets per second and connections per second between your on-premises network and your virtual network.

You can enable ExpressRoute FastPath if your virtual network gateway is Ultra Performance or ErGw3AZ.



Configure ExpressRoute FastPath

Updating an existing connection to enable FastPath

Azure PowerShell

 Copy

 Open Cloudshell

```
$connection = Get-AzVirtualNetworkGatewayConnection -Name "MyConnection" -ResourceGroupName "MyRG"
$connection.ExpressRouteGatewayBypass = $True
Set-AzVirtualNetworkGatewayConnection -VirtualNetworkGatewayConnection $connection
```

Troubleshoot ExpressRoute connection issues



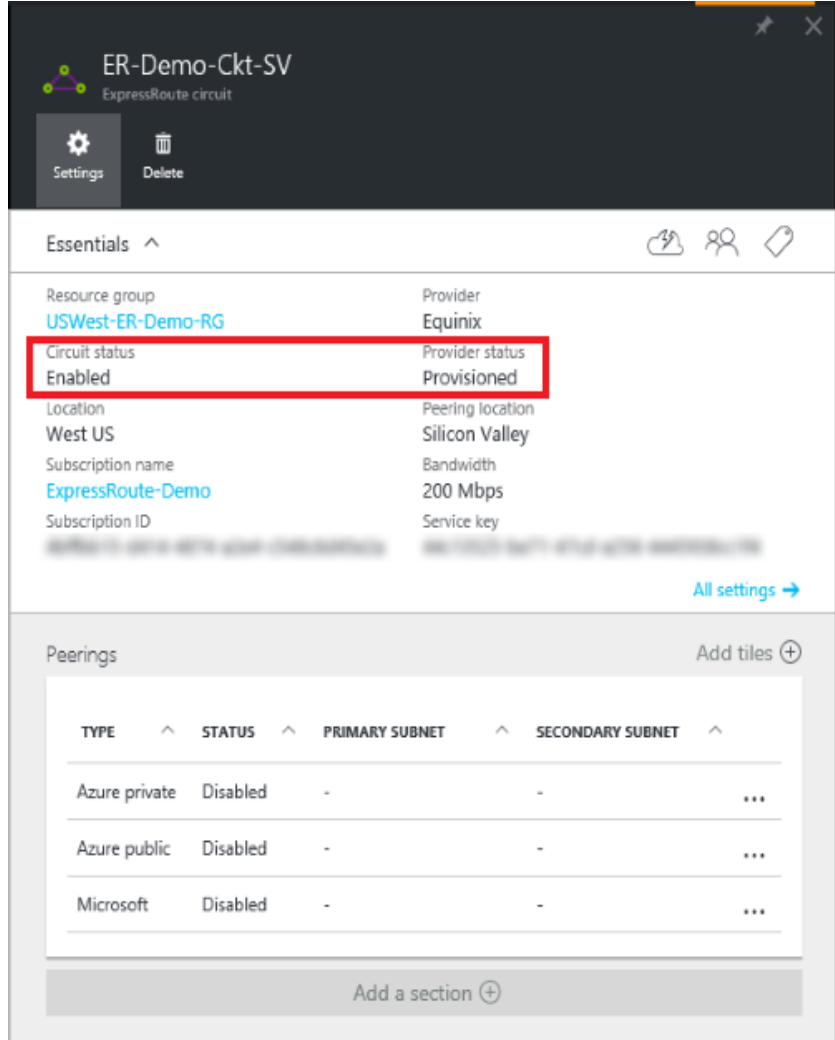
Learning Objectives - Troubleshoot ExpressRoute Connection

- Verify circuit provisioning and state through the Azure portal
- Validate Peering Configuration
- Validate Address Resolution Protocol (ARP)
- Troubleshooting network performance
- Learning Recap

Verify circuit provisioning and state

Get-AzExpressRouteCircuit -ResourceGroupName
"USWest-ER-Demo-RG" -Name "ER-Demo-Ckt-SV"

```
Name : ER-Demo-Ckt-SV
ResourceGroupName : USWest-ER-Demo-RG
Location : westus
Id : /subscriptions/*****/resourceGroups/
USWest-ER-Demo-RG/providers/*****/expressRouteCircuits/ ER-Demo-Ckt-SV
Etag : W/"#####"
ProvisioningState : Succeeded
Sku : {
  "Name": "Standard_UnlimitedData",
  "Tier": "Standard",
  "Family": "UnlimitedData"
}
CircuitProvisioningState : Enabled
ServiceProviderProvisioningState : Provisioned
ServiceProviderNotes :
ServiceProviderProperties : {
  "ServiceProviderName": "****",
  "PeeringLocation": "*****",
  "BandwidthInMbps": 200
}
ServiceKey : *****
Peerings : []
Authorizations : []
```



The screenshot shows the Azure portal interface for an ExpressRoute circuit. The top section, titled 'Essentials', displays key information about the circuit. A red box highlights the 'Circuit status' and 'Provider status' fields, both of which are 'Enabled' and 'Provisioned' respectively. Below this, a table lists various details including the resource group, location, subscription name, and bandwidth. The bottom section, titled 'Peerings', shows a table with three rows: 'Azure private', 'Azure public', and 'Microsoft', all with a status of 'Disabled'. The interface includes navigation icons for settings and deletion, and a link to 'All settings'.

TYPE	STATUS	PRIMARY SUBNET	SECONDARY SUBNET
Azure private	Disabled	-	-
Azure public	Disabled	-	-
Microsoft	Disabled	-	-

Reset a failed circuit

```
Connect-AzAccount
Get-AzSubscription
Select-AzSubscription -SubscriptionName "Replace_with_your_subscription_name"
$ckt = Get-AzExpressRouteCircuit -Name "ExpressRouteARMCircuit" -ResourceGroupName
"ExpressRouteResourceGroup"

Set-AzExpressRouteCircuit -ExpressRouteCircuit $ckt
```

The circuit should now be healthy. Open a support ticket with Microsoft support if the circuit is still in a failed state.

Validate Peering Configuration

The screenshot displays the Azure portal interface for an ExpressRoute circuit. The left-hand navigation pane is visible, with the 'Overview' tab selected and highlighted by a red rectangle. The main content area shows the circuit's details, including its resource group, status, location, subscription, and various identifiers. Below this, a 'Peering' table lists the configured peerings. The first row in the table, representing an 'Azure private' peering, is highlighted with a red rectangle.

ER-Demo-Ckt
ExpressRoute circuit

Search (Ctrl+/)

Move Delete Refresh

Resource group (change)
Demo-RG

Circuit status
Enabled

Location
East US

Subscription (change)
ExpressRoute-Lab

Subscription ID
#####-####-####-#####

Tags (change)
[Click here to add tags](#)

Provider
Equinix

Provider status
Provisioned

Peering location
Washington DC

Bandwidth
50 Mbps

Service key
#####-####-####-#####

Peering

Type	Status	Primary subnet	Secondary subnet	Last modified by	
Azure private	Provisioned	192.168.20.16/30	192.168.20.20/30	Customer	...
Azure public	Not provisioned	-	-	-	...
Microsoft	Not provisioned	-	-	-	...

Validate ARP

Address Resolution Protocol (ARP):

- layer 2 protocol defined in RFC 826.
- Used to map the Ethernet address (MAC address) with an ip address.

Title of ARP table error messages:

- On-premises MAC address show incomplete
- Microsoft entry in the ARP table

The ARP table provides a mapping of the IP address and MAC address for a particular peering. The ARP table for an ExpressRoute circuit peering provides the following information for each interface (primary and secondary):

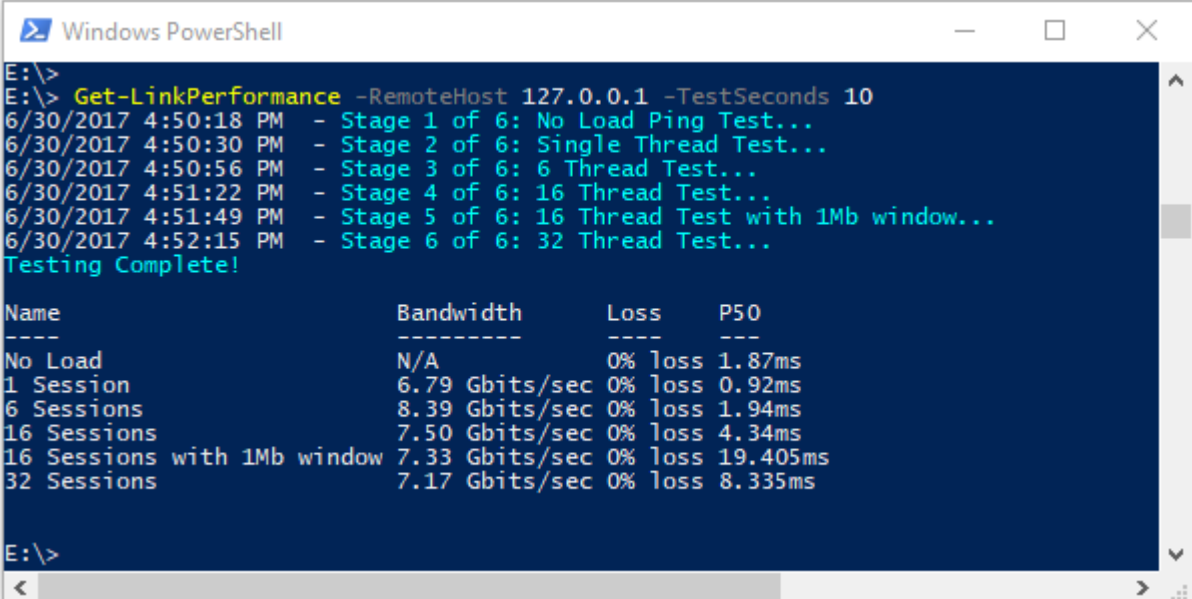
- Mapping of on-premises router interface ip address to the MAC address
- Mapping of ExpressRoute router interface ip address to the MAC address
- Age of the mapping ARP tables can help validate layer 2 configuration and troubleshooting basic layer 2 connectivity issues.

Troubleshooting network performance

Most network issues can be analyzed and isolated with Azure Network Watcher or PowerShell and CLI

To help with troubleshooting, the Azure Connectivity Toolkit (AzureCT) was developed to put some of these tools in an easy package.

These tools and methods are wrapped into a PowerShell module (AzureCT) that you can install and use



```
E:\>
E:\> Get-LinkPerformance -RemoteHost 127.0.0.1 -TestSeconds 10
6/30/2017 4:50:18 PM - Stage 1 of 6: No Load Ping Test...
6/30/2017 4:50:30 PM - Stage 2 of 6: Single Thread Test...
6/30/2017 4:50:56 PM - Stage 3 of 6: 6 Thread Test...
6/30/2017 4:51:22 PM - Stage 4 of 6: 16 Thread Test...
6/30/2017 4:51:49 PM - Stage 5 of 6: 16 Thread Test with 1Mb window...
6/30/2017 4:52:15 PM - Stage 6 of 6: 32 Thread Test...
Testing Complete!

Name                               Bandwidth      Loss      P50
----                               -
No Load                           N/A           0% loss   1.87ms
1 Session                         6.79 Gbits/sec 0% loss   0.92ms
6 Sessions                        8.39 Gbits/sec 0% loss   1.94ms
16 Sessions                       7.50 Gbits/sec 0% loss   4.34ms
16 Sessions with 1Mb window       7.33 Gbits/sec 0% loss   19.405ms
32 Sessions                       7.17 Gbits/sec 0% loss   8.335ms

E:\>
```

Lab 3:

Configure an ExpressRoute Gateway ①

Provision an ExpressRoute circuit ②

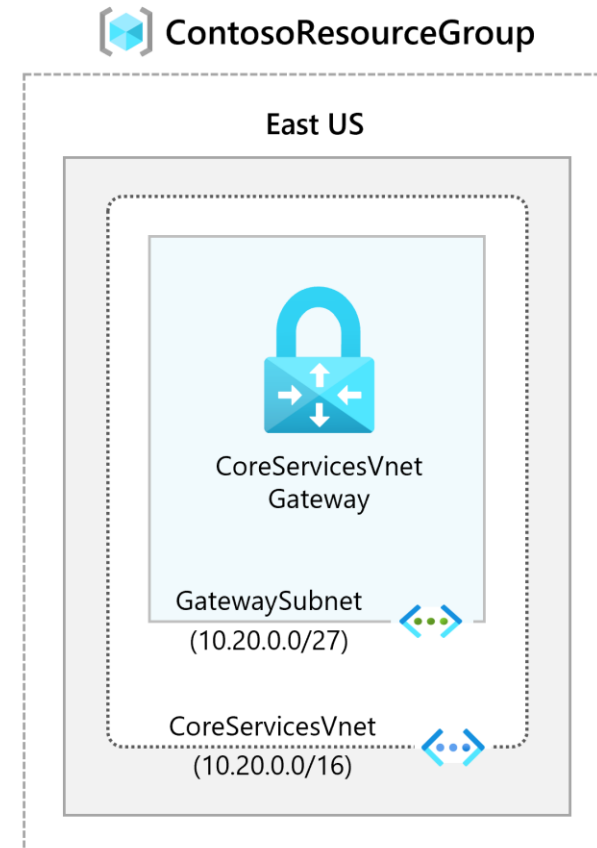


Exercise - Configure an ExpressRoute Gateway

In this exercise, you learn how to:

Task 1: Create the VNet and gateway subnet

Task 2: Create the virtual network gateway



Exercise - Provision an ExpressRoute circuit

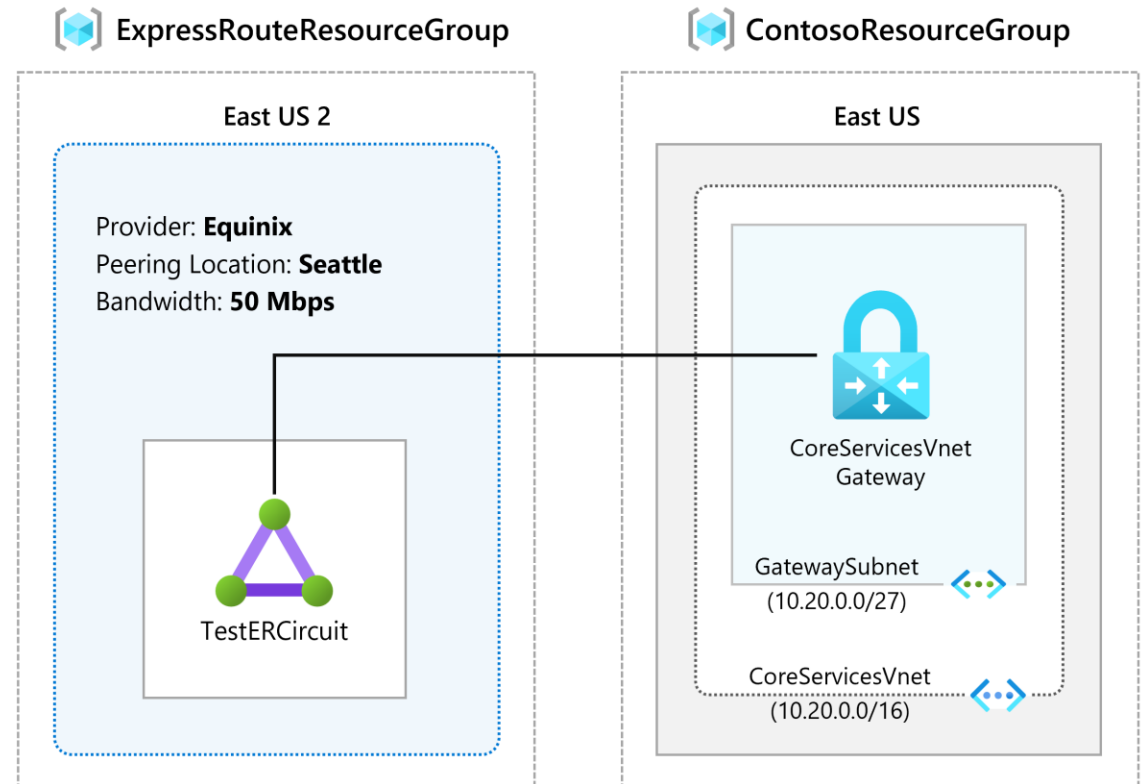


In this exercise, you learn how to:

Task 1: Create and provision an ExpressRoute circuit

Task 2: Retrieve your Service key

Task 3: Deprovisioning an ExpressRoute circuit



End of presentation

